Grade 6 Practice

Standard 6.RP.1

Name: _____ Date: _____

Computation

1. Write the ratio of 3 cats to 5 dogs. _____

3:5

2. Write the ratio of 4 red to 6 blue in simplest form. _____

2:3

3. Write the ratio of 8 to 12 in simplest form. _____

2:3

4. Write the ratio of 10 apples to 4 oranges in simplest form. _____

5:2

Word problems

5. A class has 12 boys and 18 girls. What is the ratio of boys to girls in simplest form?

Show your work and answer.

Answer: 2:3

6. Sam makes 9 of 12 free throws. What is the ratio of made shots to missed shots in simplest form?

Show your work and answer.

Answer: 3:1

Grade 6 Practice

Standard 6.RP.2

Name: _____ Date: _____

Computation

1. A car goes 150 miles in 3 hours. What is its unit rate in miles per hour? _____

50

2. 12 eggs are shared by 4 people. How many eggs per person? _____

3

3. 4 pens cost $6. What is the price per pen? _____

$1.50

4. A hiker goes 24 km in 3 hours. What is the rate in km per hour? _____

8

Word problems

5. Ana types 210 words in 5 minutes. How many words per minute is that?

Show your work and answer.

Answer: 42 words per minute

6. A printer prints 96 pages in 8 minutes. How many pages per minute is that?

Show your work and answer.

Answer: 12 pages per minute

Grade 6 Practice

Standard 6.RP.3

Name: _____ Date: _____

Computation

1. The ratio of boys to girls is 2:3. If there are 6 boys, how many girls are there? _____

9

2. A recipe uses 3 cups flour for every 2 cups sugar. How many cups of sugar go with 6 cups flour? _____

4

3. 5 pencils cost $10. What is the price of 1 pencil? _____

$2

4. On a map, 1 inch equals 50 miles. How many miles is 4 inches? _____

200

Word problems

5. Mia reads 3 pages every 5 minutes. How many minutes does it take to read 30 pages?

Show your work and answer.

Answer: 50 minutes

6. A jacket costs $40. It is on sale for 25% off. What is the sale price?

Show your work and answer.

Answer: $30

Grade 6 Practice

Standard 6.NS.1

Name: _____ Date: _____

Computation

1. Solve: $\frac{2}{3} \div \frac{1}{6} =$ _____

4

2. Solve: $\frac{3}{4} \div \frac{1}{2} =$ _____

$1\frac{1}{2}$

3. Solve: $\frac{1}{2} \div \frac{3}{4} =$ _____

$\frac{2}{3}$

4. Solve: $\frac{5}{6} \div \frac{2}{3} =$ _____

$1\frac{1}{4}$

Word problems

5. A ribbon is $\frac{3}{4}$ meter long. How many $\frac{1}{8}$-meter pieces can be cut from it?

Show your work and answer.

Answer: 6 pieces

6. How many $\frac{2}{3}$-cup servings can be made from 4 cups of rice?

Show your work and answer.

Answer: 6 servings

Grade 6 Practice

Standard 6.NS.2

Name: _____ Date: _____

Computation

1. Solve: 936 ÷ 12 = _____ **78**

2. Solve: 1,428 ÷ 14 = _____ **102**

3. Solve: 2,064 ÷ 24 = _____ **86**

4. Solve: 5,376 ÷ 32 = _____ **168**

Word problems

5. A bus holds 36 students. How many buses are needed for 900 students?

Show your work and answer.

Answer: 25 buses

6. A farm packs 1,250 eggs into cartons that hold 12 eggs. How many full cartons are there?

Show your work and answer.

Answer: 104 full cartons

Grade 6 Practice

Standard 6.NS.3

Name: _____ Date: _____

Computation

1. Solve: 3.25 + 4.8 = _____ **8.05**

2. Solve: 9.4 - 2.67 = _____ **6.73**

3. Solve: 2.5 × 3.2 = _____ **8**

4. Solve: 7.5 ÷ 2.5 = _____ **3**

Word problems

5. Sam buys 3 items that cost $2.45 each. How much does he spend in all?

Show your work and answer.

Answer: $7.35

6. A 12.6-meter rope is cut into 4 equal pieces. How long is each piece?

Show your work and answer.

Answer: 3.15 meters

Grade 6 Practice

Standard 6.NS.4

Name: _____ Date: _____

Computation

1. Find the GCF of 12 and 18. _____ **6**

2. Find the LCM of 4 and 6. _____ **12**

3. Find the GCF of 15 and 20. _____ **5**

4. Find the LCM of 3 and 5. _____ **15**

Word problems

5. Two ribbons are 24 cm and 36 cm long. They are cut into the longest equal pieces possible with none left over. How long is each piece?

Show your work and answer.

Answer: **12 cm**

6. Two lights blink every 6 seconds and every 8 seconds. After how many seconds do they blink together?

Show your work and answer.

Answer: **24 seconds**

Grade 6 Practice

Standard 6.NS.5

Name: _____ Date: _____

Computation

1. What is the opposite of 5? _____

-5

2. Start at -3 and go up 5. Where do you land? _____

2

3. Is -2 greater than -7? Write yes or no.

yes

4. Write a negative number for 40 meters below sea level. _____

-40

Word problems

5. A bank balance is -$25. Then $40 is deposited. What is the new balance?

Show your work and answer.

Answer: $15

6. Which temperature is colder, -12°F or -5°F?

Show your work and answer.

Answer: -12°F

Grade 6 Practice

Standard 6.NS.6

Name: _____ Date: _____

Computation

1. What is the opposite of -9? _____ **9**

2. What is the opposite of 0? _____ **0**

3. The point -4 is how many units from 0? _____ **4**

4. In which quadrant is the point (-3, 2)? Write the roman numeral. **II**

Word problems

5. The point (-2, 5) is in which quadrant? Write the roman numeral.

Show your work and answer.

Answer: II

6. The point (-3, -1) is in which quadrant? Write the roman numeral.

Show your work and answer.

Answer: III

Grade 6 Practice

Standard 6.NS.7

Name: _____ Date: _____

Computation

1. What is |-8|? _____

8

2. Write < or > : -3 _____ -1

<

3. Is |-6| greater than |2|? Write yes or no.

yes

4. Order from least to greatest: -4, 2, -1. _____

-4, -1, 2

Word problems

5. Sam's temperature is -7°C and another is -3°C. Which is colder?

Show your work and answer.

Answer: -7°C

6. Which is less: owing $12 or owing $5?

Show your work and answer.

Answer: owing $12

Grade 6 Practice

Standard 6.NS.8

Name: _____ Date: _____

Computation

1. Find the distance between (2, 3) and (2, 9). _____ **6**

2. Find the distance between (-4, 1) and (3, 1). _____ **7**

3. Find the distance between (0, -5) and (0, 2). _____ **7**

4. Find the distance between (-2, -3) and (-2, 4). _____ **7**

Word problems

5. A store is at (1, 2) and a park is at (1, 10). How many units apart are they?

Show your work and answer.

Answer: **8 units**

6. Two points are at (-3, 0) and (5, 0). What is the distance between them?

Show your work and answer.

Answer: **8 units**

Grade 6 Practice

Standard 6.EE.1

Name: _____ Date: _____

Computation

1. Evaluate 2^3. _____ **8**

2. Write $5 \times 5 \times 5$ using an exponent. _____ **5^3**

3. Evaluate 4^2. _____ **16**

4. Evaluate 10^3. _____ **1,000**

Word problems

5. A square has side length 6. What is its area? (Area = side²)

Show your work and answer.

Answer: 36 square units

6. A cube has edges of length 3. What is its volume? (Volume = side³)

Show your work and answer.

Answer: 27 cubic units

Grade 6 Practice

Standard 6.EE.2

Name: _____ Date: _____

Computation

1. Write an expression for 'a number n plus 7.' _____ **n + 7**

2. Write an expression for 'the product of 4 and x.' _____ **4x**

3. Evaluate 3y - 2 when y = 5. _____ **13**

4. Evaluate 2(a + 4) when a = 1. _____ **10**

Word problems

5. Cookies cost c dollars each. Write an expression for the cost of 6 cookies.

Show your work and answer.

Answer: 6c

6. Tim is t years old. His sister is 4 years older. Write an expression for her age.

Show your work and answer.

Answer: t + 4

Grade 6 Practice

Standard 6.EE.3

Name: _____ Date: _____

Computation

1. Is 2(x + 3) equal to 2x + 6? Write yes or no.

yes

2. Is 3x + 4 equal to 4 + 3x? Write yes or no.

yes

3. Simplify 4x + 2x. _____

6x

4. Simplify 5 + 2y + 3. _____

2y + 8

Word problems

5. Mia says 3(x + 2) = 3x + 2. Is she right? Write yes or no.

Show your work and answer.

Answer: **no**

6. Simplify 7a - 3a + 2. _____

Show your work and answer.

Answer: **4a + 2**

Grade 6 Practice

Standard 6.EE.4

Name: _____ Date: _____

Computation

1. Are 4 + x and x + 4 equivalent? Write yes or no. **yes**

2. Are 2y and y + y equivalent? Write yes or no. **yes**

3. Are 3x and x + 3 equivalent? Write yes or no. **no**

4. Write an expression equivalent to 5(a + 2). _____ **5a + 10**

Word problems

5. Sam says 6 + 2b and 2(3 + b) are equivalent. Is he right? Write yes or no.

Show your work and answer.

Answer: yes

6. Lee says 4 × a and a × 4 are equivalent. Is he right? Write yes or no.

Show your work and answer.

Answer: yes

Grade 6 Practice

Standard 6.EE.5

Name: _____ Date: _____

Computation

1. Is x = 3 a solution to x + 4 = 7? Write yes or no. **yes**

2. Is y = 5 a solution to 2y = 8? Write yes or no. **no**

3. Is n = 0 a solution to n + 6 = 6? Write yes or no. **yes**

4. Is a = 2 a solution to 3a = 6? Write yes or no. **yes**

Word problems

5. Is m = 4 a solution to 5m - 1 = 18? Write yes or no.

Show your work and answer.

Answer: **no**

6. Is p = 7 a solution to p - 3 = 4? Write yes or no.

Show your work and answer.

Answer: **yes**

Grade 6 Practice

Standard 6.EE.6

Name: _____ Date: _____

Computation

1. Write an expression for the number of cookies in n boxes, with 12 cookies per box. _____

12n

2. Write an expression for 'x minus 3.' _____

x − 3

3. Write an expression for 'twice a number n plus 5.' _____

2n + 5

4. Write an expression for 'half of t.' _____

t/2

Word problems

5. A taxi charges $3 plus $2 per mile. Write an expression for the cost of m miles.

Show your work and answer.

Answer: **3 + 2m**

6. A bag holds b apples. Write an expression for 5 bags plus 4 loose apples.

Show your work and answer.

Answer: **5b + 4**

Grade 6 Practice

Standard 6.EE.7

Name: _____ Date: _____

Computation

1. Solve: x + 5 = 12. x = _____

7

2. Solve: 3y = 18. y = _____

6

3. Solve: n - 4 = 10. n = _____

14

4. Solve: k/2 = 7. k = _____

14

Word problems

5. Sam has x dollars. He earns 9 more and now has 20 dollars. Find x.

Show your work and answer.

Answer: 11 dollars

6. Each box holds b cards. 4 boxes hold 28 cards. Find b.

Show your work and answer.

Answer: 7 cards

Grade 6 Practice

Standard 6.EE.8

Name: _____ Date: _____

Computation

1. Is 6 a solution to x > 5? Write yes or no. **yes**

2. Is 3 a solution to x < 3? Write yes or no. **no**

3. Which number satisfies x ≥ 4: 2, 3, or 4? Write the number. **4**

4. Write an inequality for 'y is less than 10.' _____ **y < 10**

Word problems

5. Max needs at least 25 points. Write an inequality for the points p he needs.

Show your work and answer.

Answer: **p ≥ 25**

6. Jo has at most $15 to spend. Write an inequality for the amount s she can spend.

Show your work and answer.

Answer: **s ≤ 15**

Grade 6 Practice

Standard 6.EE.9

Name: _____ Date: _____

Computation

1. In d = 5t, which variable is independent? Write d or t.

t

2. In C = 3n, which variable is dependent? Write C or n.

C

3. In the cost c = 4p, p is the _____ variable.

independent

4. In y = x + 2, which variable is dependent? Write y or x.

y

Word problems

5. A plant grows 2 cm per day. Its height h after d days is h = 2d. Which variable is independent?

Show your work and answer.

Answer: d

6. Pay P depends on hours h worked: P = 12h. Which variable is dependent?

Show your work and answer.

Answer: P

Grade 6 Practice

Standard 6.G.1

Name: _____ Date: _____

Computation

1. A triangle has base 6 and height 4. What is its area? _____ **12**

2. A parallelogram has base 5 and height 3. What is its area? _____ **15**

3. A triangle has base 10 and height 3. What is its area? _____ **15**

4. A trapezoid has bases 4 and 6 and height 3. What is its area? _____ **15**

Word problems

5. A triangular garden has a base of 8 m and a height of 5 m. What is its area?

Show your work and answer.

Answer: **20 square meters**

6. A parallelogram has base 7 and height 2. What is its area?

Show your work and answer.

Answer: **14**

Grade 6 Practice

Standard 6.G.2

Name: _____ Date: _____

Computation

1. A rectangular prism is 4 by 3 by 2. What is its volume? _____

24

2. A prism has a base area of 10 and a height of 5. What is its volume?

_____

50

3. A cube has side length 3. What is its volume? _____

27

4. A prism has a base area of 6 and a height of 4. What is its volume?

_____

24

Word problems

5. A box is 5 by 2 by 3 units. What is its volume?

Show your work and answer.

Answer: 30 cubic units

6. A fish tank has a base of 4 by 6 feet and a height of 2 feet. What is its volume?

Show your work and answer.

Answer: 48 cubic feet

Grade 6 Practice

Standard 6.G.3

Name: _____ Date: _____

Computation

1. A triangle has vertices (0, 0), (4, 0), and (4, 3). What is its area? _____

6

2. A rectangle has corners (1, 1), (5, 1), (5, 4), and (1, 4). What is its area? _____

12

3. What is the length of the segment from (2, 1) to (2, 7)? _____

6

4. What is the length of the segment from (-1, 3) to (4, 3)? _____

5

Word problems

5. A rectangle has vertices (0, 0), (6, 0), (6, 2), and (0, 2). What is its area?

Show your work and answer.

Answer: **12 square units**

6. Find the distance between the points (1, 2) and (1, 9).

Show your work and answer.

Answer: **7 units**

Grade 6 Practice

Standard 6.G.4

Name: _____ Date: _____

Computation

1. How many faces does a cube have? _____ **6**

2. How many edges does a rectangular prism have? _____ **12**

3. How many faces does a triangular prism have? _____ **5**

4. How many triangular faces does a square pyramid have? _____ **4**

Word problems

5. How many squares are in a net of a cube?

Show your work and answer.

Answer: 6 squares

6. A net has 2 triangles and 3 rectangles. Which solid does it make?

Show your work and answer.

Answer: triangular prism

Grade 6 Practice

Standard 6.SP.1

Name: _____ Date: _____

Computation

1. Which is a statistical question? Write A or B. A) How tall are students in our class? B) What is 2 + 2?

A

2. Is 'What is your favorite color?' a statistical question? Write yes or no.

yes

3. Is 'How many pets do students in our class have?' a statistical question? Write yes or no.

yes

4. Is 'What is 5 × 5?' a statistical question? Write yes or no.

no

Word problems

5. Is 'Who is the tallest person in the room?' a statistical question? Write yes or no.

Show your work and answer.

Answer: no

6. Is 'How long is a typical bus ride for our grade?' a statistical question? Write yes or no.

Show your work and answer.

Answer: yes

Grade 6 Practice

Standard 6.SP.2

Name: _____ Date: _____

Computation

1. Data: 2, 4, 6, 8, 10. What is the median? _____ **6**

2. Data: 3, 3, 5, 9. What is the mode? _____ **3**

3. Data: 4, 9, 12, 20. What is the range? _____ **16**

4. Data: 1, 2, 3, 4. What is the median? _____ **2.5**

Word problems

5. Test scores are 70, 80, and 90. What is the mean?

Show your work and answer.

Answer: **80**

6. Heights are 50, 52, 54, and 56 inches. What is the range?

Show your work and answer.

Answer: **6 inches**

Grade 6 Practice

Standard 6.SP.3

Name: _____ Date: _____

Computation

1. Find the mean of 2, 4, and 6. _____ **4**

2. Find the median of 1, 3, 5, and 7. _____ **4**

3. Find the mode of 2, 2, 5, and 6. _____ **2**

4. Find the mean of 10, 20, 30, and 40. _____ **25**

Word problems

5. Sam's steps for 5 days were 3, 5, 7, 5, and 5. What is the mode?

Show your work and answer.

Answer: 5 steps

6. Sam's test scores are 80, 90, and 100. What is the mean?

Show your work and answer.

Answer: 90

Grade 6 Practice

Standard 6.SP.4

Name: _____ Date: _____

Computation

1. A dot plot shows 3 dots above the number 2. How many data points are there? _____

3

2. A histogram has 5 data points in the 0–4 bar. How many data points are there? _____

5

3. A dot plot has 2 dots at 1 and 3 dots at 2. How many data points in all? _____

5

4. A histogram has 6 data points in 10–20 and 4 in 20–30. How many data points in all? _____

10

Word problems

5. A dot plot shows 4 students with 2 pets. How many students are in the plot?

Show your work and answer.

Answer: 4 students

6. A histogram has 8 data points in the 0–5 group and 2 in the 5–10 group. How many data points in all?

Show your work and answer.

Answer: 10

Grade 6 Practice

Standard 6.SP.5

Name: _____ Date: _____

Computation

1. A data set has 8 values. How many values does it have? _____

8

2. Find the mean of 5, 7, and 9. _____

7

3. Find the range of 3, 8, and 10. _____

7

4. The measure of spread found by subtracting the minimum from the maximum is called the _____.

range

Word problems

5. A data set has 6 values. The maximum is 20 and the minimum is 4. What is the range?

Show your work and answer.

Answer: **16**

6. Four values have a mean of 12. What is the sum of the four values?

Show your work and answer.

Answer: **48**